The Islamia University of Bahawalpur
Department of Information Technology

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E-Commerce Application Development

Chapter
13

Web Payment Systems

Part IV — Money and Identity

Taught in	Week 10
Laboratory	A sandbox gateway, and what happens when it times out
Assessment	**Project proposal**

This is a class extract. It is *Chapter 13* of the unified handout, printed on its own so that a week's reading is a week's reading. Page numbers restart at 1 and are local to this extract; chapter, figure and table numbers are the ones the full volume gives them, so a reference you follow here lands in the same place there.

Cross-references to other chapters print as their numbers in the full volume. For the whole course in one file — front matter, the HEC tracing table, all twenty-three chapters and all seven appendices — see `ECAD_Fall2026_Handout.pdf`.

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Chapter 13

Web Payment Systems

Learning Outcomes

By the end of this chapter you will be able to:

- **Name** the parties to an online payment and **state** what each one does and is paid for.
- **Distinguish** the payment instruments available to a Pakistani shop, and **compare** them on cost, settlement time and risk.
- **Explain** who carries the loss when a payment goes wrong, for each instrument.
- **Compute** the true cost of cash on delivery, including the returns it causes.
- **Select** a payment mix for a stated shop, and **defend** the choice against the alternatives.

Before You Start

Chapter 10: the order state machine, and that an order exists before it is paid for. Chapter 11: order totals, so that percentages have something to be percentages of.

Key Terms in This Chapter

payment instrument • card scheme • issuer • acquirer • payment gateway • payment service provider • merchant discount rate • settlement • chargeback • authorisation • capture • mobile wallet • cash on delivery • return-to-origin • Raast • PayPak

13.1 What Happens When Somebody Pays

A customer clicking **Pay** starts a conversation between five parties, four of which the shop never sees. Knowing who they are is not trivia: it is how you work out who to telephone when money is missing, and who will be out of pocket if it stays missing.

The five parties

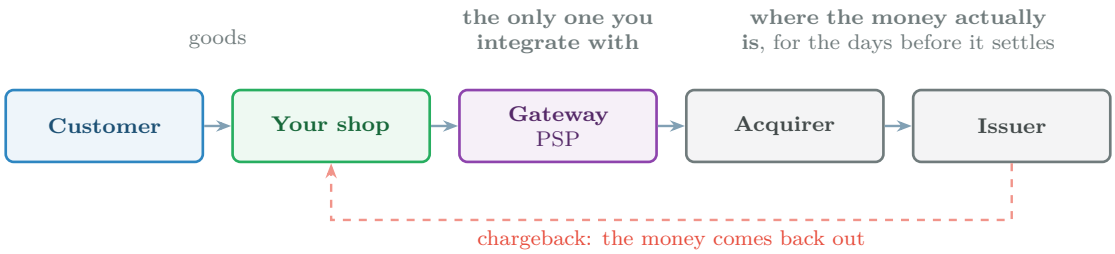
The **cardholder** or payer. Your customer.

The **issuer** — the bank that gave them the card or holds their wallet. It decides whether to approve the payment, and it is the party the customer complains to.

The **scheme** — Visa, Mastercard, PayPak — which operates the rules and the network between banks. It writes the rulebook that decides disputes.

The **acquirer** — the bank that holds the *shop's* account and accepts the payment on its behalf. It carries the risk that the shop fails to deliver, which is why it asks about your business before accepting you.

The **gateway** or **payment service provider** — the software company between the shop and the acquirer. It is the only one of the five you will integrate with, and the only one whose documentation you will read.



You pay all four. The merchant discount rate is a single percentage that is then divided between scheme, acquirer and gateway — which is why nobody will quote you its parts.

Figure 13.1: The parties to a card payment, and the direction a chargeback travels. The shop sits at one end of a chain it cannot see into, which is the central fact about accepting cards.

Authorisation is not payment

Two separate events, and confusing them is how shops ship goods they are never paid for.

Authorisation asks the issuer "would this payment succeed?" and reserves the amount. It is instant, it is reversible, and it is not money.

Capture claims the reserved amount. Only now is the shop owed anything.

Settlement is the acquirer actually paying the shop, typically days later and net of fees.

A gateway that reports "success" may mean any of the three. Read its documentation for which, because Chapter 14's entire difficulty is that your order status must mean the same thing as the gateway's.

13.2 The Instruments a Pakistani Shop Can Actually Offer

The platform ships no online payment method at all

Checked rather than assumed. The payment extensions in the box are `bank_transfer`, `cheque`, `cod` and `free_checkout` — **four methods, none of which takes money online**. Every one of them records an intention and leaves the shop to collect afterwards.

There is no card gateway, no wallet, nothing. Every real payment integration is a third-party extension, bought or written, and Chapter 14 is the chapter that deals with that. A plan that assumed "the platform handles payments" has already failed, and it is better to discover that here than in week fourteen.

Instrument	Typical cost	Settles	Who loses if it goes wrong	Main difficulty
Card (Visa, Mastercard)	2–3%	days	**the shop**, via chargeback	disputes you cannot win
PayPak card	lower	days	the shop	domestic acceptance only
Mobile wallet (JazzCash, Easypaisa)	1–2%	days	mostly the shop	reconciliation

Instrument	Typical cost	Settles	Who loses if it goes wrong	Main difficulty
Bank transfer / Raast	very low	near instant	effectively nobody	confirming which order was paid
Cash on delivery	1–3% plus returns	on remittance	**the shop**, heavily	refusal at the door

Rates are indicative and negotiated. Any figure here is a starting point for a conversation with a provider, not a price. What does not change is the *shape*: online card acceptance is the most expensive and carries the dispute risk; bank transfer is the cheapest and carries an operational burden instead; cash on delivery looks cheap and is not.

13.3 Cash on Delivery Is Not Free

Cash on delivery dominates Pakistani e-commerce for a reason that has nothing to do with technology: it asks the customer to trust nobody. They pay when the goods are in their hand. A shop that refuses COD is refusing most of its market.

It is also the most expensive instrument on the list, and the cost is invisible because it does not arrive as a percentage.

Worked Example:

A shop sells an item for **Rs 3,000** that costs it **Rs 2,100** — a 30% gross margin, Rs 900 per order.

The visible cost. The courier charges a COD service fee, say **2%** of the collected amount: $0.02 \times 3{,}000 = Rs\ 60$.

The invisible cost. Some proportion of COD orders are refused at the door — the customer is out, has changed their mind, or ordered on an impulse that has passed. Call it **20%**, which is not a pessimistic figure for unverified COD orders. A refused parcel is **return-to-origin**: the shop pays delivery *and* return, say **Rs 250** each way, and gets the item back with nothing to show.

Per 100 orders:

- 80 delivered: $80 \times (900 - 60 - 250) = 80 \times 590 = Rs\ 47{,}200$
- 20 refused: $20 \times (-250 - 250) = -Rs\ 10{,}000$
- **Net: Rs 37,200**, against Rs 72,000 of gross margin on the same 100 orders if they were prepaid and delivery were free.

The effective cost of COD here is 48% of gross margin — against 2–3% for a card. The refusal rate, not the service fee, is almost all of it: the 20 refused orders cost Rs 10,000 directly and also consumed the Rs 20,000 of margin they would have produced.

The lever that matters is therefore the refusal rate, not the fee. Dropping refusals from 20% to 10% is worth more than negotiating the courier fee to zero — which is why shops that take COD invest in confirming orders by telephone before dispatch, and why an order confirmation step that feels like friction is, here, the single most profitable thing in the pipeline.

What this says about the "frictionless checkout" advice

Most writing about e-commerce checkout comes from markets where payment is taken before dispatch. There, every removed step is pure gain, and the advice is sound.

Where the shop is paid on delivery, an unverified order is a liability rather than an asset, and a checkout optimised purely for completion produces more of them. **Advice imported without its assumptions is how a shop loses money by following best practice.** Chapter 8 made the same point about page weight; this is the version that costs real money.

13.4 The Local Rails

What a Pakistani shop is actually plugged into

- **State Bank of Pakistan** authorises and supervises the operators and providers. A gateway you integrate with should be an authorised one, and asking for that authorisation is a reasonable first question.
- **1LINK** operates the domestic switch connecting the banks, and **PayPak** is the domestic card scheme that runs over it. A PayPak card is cheaper to accept and works only in Pakistan.
- **Raast**, the State Bank's instant payment system, settles bank to bank in seconds at negligible cost. Its person-to-merchant use is the development most likely to change the arithmetic in this chapter, because it removes both the card fee and the COD refusal problem at once.
- **JazzCash** and **Easypaisa** are the mobile wallets, each with a merchant gateway, and between them they reach customers who have no card and no bank account — which in this market is the majority.

Two absences worth knowing. PayPal does not operate for merchants in Pakistan, and Stripe does not support Pakistan-registered businesses. A great deal of the integration tutorial material you will find online assumes one or the other, and none of it applies. Students regularly lose a week to this.

Dated facts, and how to treat them

Everything in this section is a fact about a particular year. Rates are negotiated, operators are authorised and de-authorised, and Raast's merchant rollout is still changing what is possible.

The durable part is the set of questions: who is authorised, what does it cost as a percentage and as a fixed fee, how long until the money is in the shop's account, who decides a dispute, and what proportion of customers can actually use it. Ask those five of any provider in any year and the answer will be current. Memorising this year's rates will not help you in 2029; knowing that settlement time and dispute liability are negotiable terms will.

Common Pitfalls

- **Treating an authorisation as a payment** and dispatching goods against it.
- **Quoting a client a card fee without the chargeback risk.** The 2–3% is the predictable part; the disputes are not.
- **Costing cash on delivery at the courier's service fee.** The refusal rate is most of the cost and appears in no invoice.
- **Offering only cards in a market where most customers have none.**
- **Following a payment integration tutorial written for PayPal or Stripe.** Neither is available to you here.
- **Assuming the platform can take payments out of the box.** Its four methods all collect later.

- **Letting the customer choose the currency without deciding who carries the exchange risk.**

Try It Yourself: Cost your own shop's payment mix

```
# --- 1. See what you actually have ----------------------------------
cd code
docker compose exec shop sh -c \
  'ls extension/opencart/admin/controller/payment/'
# Four files. Which of them takes money online? Write the answer down.

# --- 2. Enable bank transfer and read what it does ------------------
# Admin: Extensions > Extensions > Payments > Bank Transfer > Install.
# Place an order with it. What does the customer see, and what does
# the order status become?

# --- 3. Compare that with cash on delivery --------------------------
# Place one of each and compare the two orders' status histories.

# --- 4. Build the COD model for YOUR product ------------------------
# Take one product in your shop. Find its price. Assume a cost of 70%
# of price, Rs 250 each way for delivery, a 2% COD fee, and a 20%
# refusal rate. Compute net margin per 100 orders.

# --- 5. Find the break-even refusal rate ----------------------------
# At what refusal rate does COD earn the shop nothing at all?
# Solve it, do not guess.

# --- 6. Price the alternative ---------------------------------------
# Repeat step 4 for a 2.5% card fee, zero refusals, and the same
# delivery cost one way only. Which is better, and by how much?

# --- 7. Decide, and write the decision down -------------------------
# One paragraph: which instruments your shop will offer, in what
# order on the checkout page, and what you will measure to find out
# whether you were right.
```

What to notice. Step 5 surprises people. The break-even refusal rate is lower than most shops' actual refusal rate, which means a shop can run on COD for a year, grow its order count every month, and lose money on every single one.

Checkpoint — Test Yourself

1. A gateway reports "success" and your shop marks the order Processing and dispatches it. Describe two distinct ways the shop can end up having shipped goods it is never paid for, and say what you would check in the gateway's documentation to rule each one out.
2. Using the worked model, compute the net margin per 100 orders if the refusal rate falls from 20% to 10%. State the improvement as a percentage, and compare it with what the shop would gain by negotiating the courier's COD fee from 2% to 0%.
3. A client wants to accept international cards for customers abroad. List, in order, the four things you would establish before writing any code, and say which of them is most likely to stop the project.

Chapter Summary

- Five parties: customer, issuer, scheme, acquirer, gateway. You integrate with one and pay all of them through a single merchant discount rate whose parts nobody will itemise.

- Authorisation, capture and settlement are three different events. A gateway's "success" means one of them, and your order status must mean the same thing.
- The platform ships **no online payment method** — bank transfer, cheque, cash on delivery and free checkout all collect later. Every gateway is a third-party extension.
- Instruments differ less in price than in who carries the loss. Cards put the dispute risk on the shop; bank transfer puts an operational burden on it instead.
- Worked: cash on delivery at a 20% refusal rate costs 48% of gross margin, almost all of it from refusals rather than from the service fee. The refusal rate is the lever; the fee is not.
- Checkout advice written for prepaid markets does not transfer. Where payment is on delivery, an unverified order is a liability and confirming it is the most profitable step in the pipeline.
- Locally: SBP authorises, 1LINK switches, PayPak is the domestic scheme, Raast settles instantly, and the wallets reach customers with no card. PayPal and Stripe are not available to you.
- Rates and operators date; the five questions — authorised, cost, settlement time, who decides disputes, who can use it — do not.

Review Questions

1. **(MCQ)** Which party carries the loss when a card payment is successfully disputed? (a) the issuer; (b) the scheme; (c) the acquirer; (d) the shop.

2. **(MCQ)** How many of the payment extensions shipped with the platform take money online? (a) none; (b) one; (c) two; (d) all four.

3. **(Short)** Distinguish authorisation, capture and settlement, and say which one a shop should require before dispatching goods.

4. **(Short)** Explain why a 2% COD service fee understates the cost of cash on delivery, using one number from the worked model.

5. **(Short)** Give two reasons a Pakistani shop might prefer PayPak acceptance to Visa and Mastercard, and one reason it cannot rely on it alone.

6. **(Applied)** A shop takes 600 orders a month at an average value of Rs 4,000 and a 28% gross margin, all on cash on delivery with an 18% refusal rate and Rs 300 of delivery cost each way. Model the effect of introducing prepayment with a 2.5% fee that 35% of customers adopt, and state the monthly difference, showing every step.

7. **(Applied)** You are asked to decide whether to add a mobile wallet to a shop that currently takes only COD. Set out what you would need from the provider, what you would change in the shop, what you would measure for how long, and what result would make you remove it again.